OligoR: workflow for oligonucleotide HDX/MS data analysis

ARNA

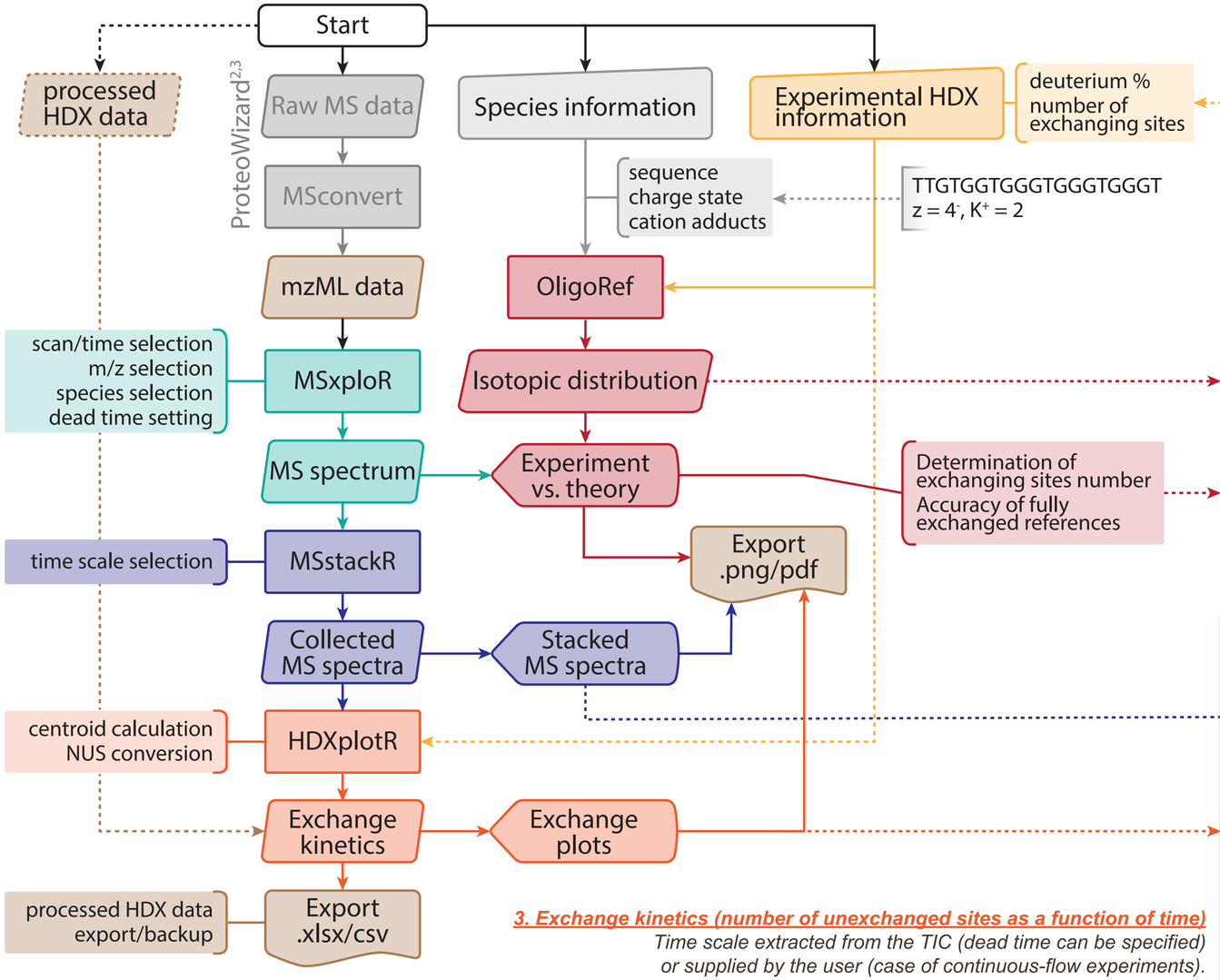
- Many programs exist to specifically analyze protein HDX/MS data
- OligoR is a software dedicated to **oligonucleotide native (HDX/)MS**
- From raw **data treatment** to processed data **visualization**
- Coded in R, using Shiny for interactivity purposes

Example data acquired on a Thermo Exactive mass spectrometer, using the T30177-TT aptamer (HIV-1 integrase enzyme inhibitor) as example[1]

Interface, divided in 4 modules:
- **OligoRef**: compares theoretical isotopic distributions to experimental data
- **MSxploR**: imports MS files, combines scans and selects m/z ranges
- **MSstackR**: generates stacked-spectrum figures of selected time points
- **HDXplotR**: calculates *m/z* centroids and numbers of unexchanged sites (*NUS*)

Data management
- **Open MS file formats** (mzML, mzXML) input, using the mzR package[2-4]
- **Data traceability** ensured by associating raw data filenames, selected scan and m/z ranges to processed data point
- Data analysis process fully **reproducible**
- Processed data can be **saved in cross-platform formats** (.csv, .xlsx, copy to clipboard) for external use
- Processed data can be **reimported** along newly processed data

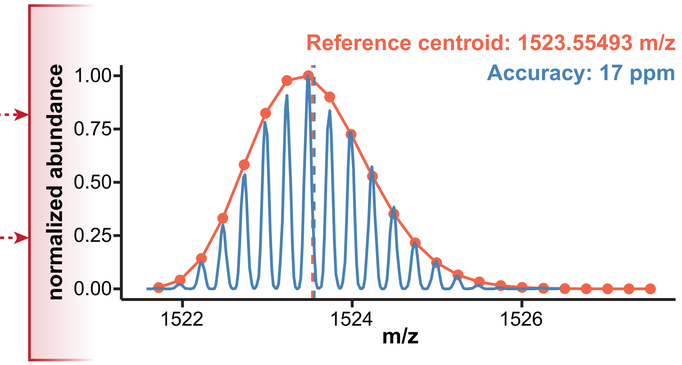
Publication-ready figures
- **Customizable**
- **Exportable** as png or pdf files

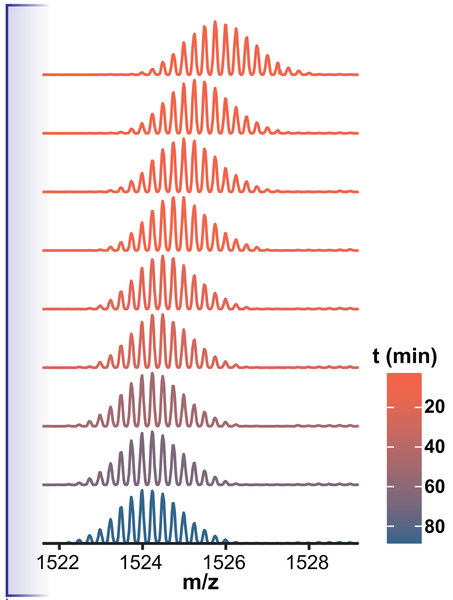


1. Theoretical and experimental isotopic distributions of references

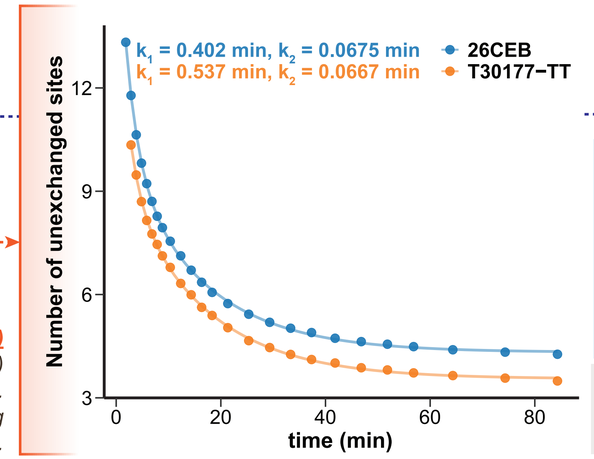
Theoretical distribution *computed by FFT from species and HDX information, at specified %D (here, 9% for exchanged reference)*

Experimental distribution *generated by user by combining MS scans from fully-exchanged reference sample*





2. Stacked MS spectra*, generated automatically from all user-selected time points. The color scale can be set to linear or logarithmic to accommodate large time range.*



3. Exchange kinetics (number of unexchanged sites as a function of time)

Time scale extracted from the TIC (dead time can be specified) or supplied by the user (case of continuous-flow experiments). Several species can be compared, from the same dataset or by re-importing already processed data (here the results for the human minisatellite 26CEB were imported).

- OligoR is an open-source software dedicated to **native (HDX/)MS data of oligonucleotides**
- Semi-automated process from raw data to **customizable figures**: fast and user-friendly
- **Traceable data** and **reproducible process**

To see OligoR in action, watch the video!

Follow the project on https://github.com/EricLarG4/OligoR

1. Largy,E., Gabelica,V. (**2020**) *Anal. Chem.*, **92**, 4402
2. Chambers,M.C., Maclean,B., Burke,R., *et al.* (**2012**) *Nat. Biotechnol.*, **30**, 918
3. Kessner,D., Chambers,M., Burke,R., *et al.* (**2008**) *Bioinformatics*, **24**, 2534
4. Martens,L., Chambers,M., Sturm,M., *et al.* (**2011**) *Mol. Cell. Proteomics*, **10**, 1